

Khondker Salman Sayeed

Software Engineer @ IQVIA

Website  Github  LinkedIn  Scholar 

WORK EXPERIENCE

Software Engineer, **IQVIA**

JULY 2024 - PRESENT

- **Architected and developed agentic AI components** for IQVIA's AI Assistant, enabling intelligent automation and decision-making, reducing manual data retrieval by 80% through **real-time RAG** and **ReAct Agents**.
- **Built and optimized a Mixture-of-Agents AI pipeline** using **LangChain**, **LangGraph**, and internal tools, enabling interactive, data-driven responses from enterprise-scale datasets in **Snowflake**, **Bedrock**, and **Firebolt**, improving accuracy by 40%.
- **Designed and deployed scalable microservices with FastAPI**, and **.NET** in a **multi-cloud (AWS & Azure) environment**, improving system throughput by 50%, ensuring seamless delivery of healthcare-grade AI solutions to thousands of end-users.
- **Maintaining and optimizing production ML pipelines**, leveraging **AWS Bedrock**, **SageMaker**, and **Hugging Face**, reducing model deployment time by 60% and ensuring 99.9% uptime.
- **Developing and fine-tuning** IQVIA's in-house domain-specific **LLMs**, integrating **Transformer** architectures, **LoRA**, and **quantization** techniques, boosting analytical efficiency by 35% for life sciences professionals.
- **Automating CI/CD** with **GitLab CI/CD**, reducing release cycles by 35%.

Software Engineer (Part-time), **ResTechAI**

JULY 2024 - MARCH 2025

- **Leading development of a drag-and-drop ML platform**, cutting model deployment time by 50%, enabling seamless AI pipeline creation.
- Developing agentic ML pipelines with **LangChain**, **LangGraph**, and **LlamaIndex**, optimizing inference efficiency by 40% with **ChromaDB**, **PostgreSQL** integrations.
- **Architecting full-stack microservices** using **FastAPI**, **TypeScript**, **Express.js**, **React**, **Docker**, and **Kubernetes**, improving API performance by 30%.
- **Customizing and integrating open-source AI tools**, enhancing automation and usability with **Ollama**, **LiteLLM**, **AnythingLLM**, **Windmill**, and **Flowise**.
- **Optimizing ETL workflows** for structured/unstructured data sources, reducing processing latency by 35%.

Research Assistant, **CSE BUET NLP Lab**

APRIL 2023 - JUNE 2024

- **Developed a benchmark illusion dataset** to evaluate the visual understanding of **Vision-Language Models (VLMs)**, published at **COLM'24**.
- **Conducting research on multimodal deep learning and VLMs** for an undergraduate thesis, focusing on **text-image understanding and fusion techniques**.
- **Curated a large-scale dataset** of newspaper articles, images, and summaries to enhance **abstractive summarization**, integrating image data for improved VLM performance.

EDUCATIONAL BACKGROUND

Bachelor of Science in Computer Science and Engineering

Bangladesh University of Engineering and Technology

APRIL 2019 - JULY 2024

- CGPA: 3.9/4.0 (Top 15% of my class)
- University **Merit Scholarship** Recipient (4/6 Semesters)
- University **Dean's List** Scholarship Recipient (All Semesters)

CONTACT

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PROGRAMMING LANGUAGES

- Python
- C/ C++ / CUDA C
- Java
- Javascript / Typescript
- Assembly
- Bash

TECHNOLOGIES & TOOLS

- **Backend Frameworks:** Express.js, Next.js, FastAPI, Flask
- **Frontend:** React, Next.js, TailwindCSS, Material UI, Vite, Jinja, Pug
- **Databases & Storage:** MySQL, PostgreSQL, MongoDB, OracleDB, SQLite, Redis
- **DevOps & Cloud:** AWS (EC2, S3, Lambda, Sagemaker, Bedrock, OpenSearch), Google Cloud (GCP), Docker, Docker Compose, Kubernetes, GitHub Actions, GitLab CI/CD, Nginx
- **Testing & QA:** Jest, PyTest, Postman, Selenium, Playwright, Swagger
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, Scikit-Learn, CUDA, Hugging Face, OpenCV, ONNX
- **Tools & Workflow:** Git, Jira, Confluence, WebSockets, RabbitMQ, NATS, gRPC, REST

RESEARCH PUBLICATIONS

IllusionVQA: A Challenging Optical Illusion Dataset for Vision Language Models

Haz Sameen Shahgir*, Khondker Salman Sayeed*, Abhik Bhattacharjee, Ahmad Wasi Uddin, Yue Dong, Rifat Shahriyar
[Paper] [Website] [Code]

COLM 2024 & SoCal NLP Symposium 2024

"RNA-DCGen: Dual Constrained RNA Sequence Generation with LLM-Attack

Haz Sameen Shahgir*, Md Rownok Zahan Ratul*, Md Toki Tahmid*, Khondker Salman Sayeed, Atif Rahman
[Paper]

Machine Learning in Structural Biology (MLSB) Workshop, NeurIPS 2024

Too Late to Train, Too Early To Use? A Study on Necessity and Viability of Low-Resource Bengali LLMs

Tamzeed Mahfuz*, Satak Kumar Dey*, Ruwad Naswan*, Hasnaen Adil, Khondker Salman Sayeed, Haz Sameen Shahgir
[Paper]

COLING 2025

PROJECTS

-  **Ticketing** — *A microservices ticketing webapp*
 - Consisting of 6 decoupled services through an event-driven architecture using NATS event bus.
 - Backend: TypeScript, Node.js, Express.js.
 - Frontend: Next.js, TailwindCSS
 - Storage: MongoDB, Redis
 - Tools: NATS Event bus, Kubernetes, Ingress-Nginx, Scaffold, GitHub Actions CI/CD
 - Testing: Jest
 - Cloud: GKE (Google Kubernetes Engine)
-  **Sub-C Compiler** — *A compiler for a subset of the C programming language*
 - Generates x86 assembly, shows various lexical and syntax errors on erroneous source programs
 - Tools: Flex (Lexer), GNU Bison (Parser)
 - Language: C, C++
-  **Mobile-Doc** — *A monolithic telemedicine web application*
 - Backend: FastAPI
 - Frontend: React, Material UI, TailwindCSS
 - Storage: MongoDB, Redis, BigQuery
 - Tools: GitHub CI/CD, Google Cloud Build
 - Testing: PyTest
 - Cloud: Google Cloud (Backend), Netlify (Frontend)
-  **Suntech** — *A monorepo e-commerce web application*
 - Backend: Flask
 - Frontend: Jinja2, HTML, CSS, Javascript
 - Storage: OracleDB
-  **Ray-Tracing** — *A rendering pipeline implementing ray tracing on OpenGL frames*
 - Algorithms: Phong Lighting Model, Recursive Reflections
 - Language: C++
 - Tools: OpenGL
-  **Enhanced xv6 OS** — *Copy-On-Write and Memory Page Replacement in the xv6 OS*
 - Language: C
 - System Calls, Process Schedulers, Page Replacement
-  **twitt3r** — *A Twitter clone using the t3 stack*
 - T3 Stack (Typescript, trpc, TailwindCSS)
 - Backend: TypeScript, Next.js
 - Frontend: Next.js, TailwindCSS
 - Storage: PlanetScale, PrismaORM
 - Cloud: Vercel
-  **DocumentQA** — *A langchain LLM document chatbot*
 - Python Chainlit application
 - Storage: ChromaDB
 - Tools: LangChain, OpenAI
 - Database: ChromaDB Vector Database

REFERENCES

Srikanth Sankaran Iyer Associate Director, IQVIA srikanth.iyer@iqvia.com Relation: Manager	Rifat Shahriyar Professor at CSE, BUET rifat@cse.buet.ac.bd Relation: Thesis Supervisor
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AWARDS AND COMPETITIONS

IEEE Upsilon Pi Epsilon Honor Society Award 2023

Winner

Awarded by IEEE Computer Society
MARCH 2024

- Received award based on academic excellence and extracurricular activities in the field of computing (Awarded to only 4 applicants internationally)
- Received a \$1000 scholarship

IEEE Video and Image Processing Cup 2023

Champion

Hosted by IEEE Signal Processing Society, Georgia Institute of Technology
ICIP 2023, Kuala Lumpur, Malaysia
AUGUST - OCTOBER 2023

- Task: Biomarker Prediction from retina OCT scans
- Model: MaxViT, EVA02-ViT, InceptionNet, ResNet
- Competition Page: [2023 IEEE SPS Video and Image Processing \(VIP\) Cup](#)
- Team Name: Synapse
- Paper: [\[arXiv\]](#)

DLsprint 2022

Champion

Hosted by CSE BUET, Bengali.AI and Optimizely
JULY - SEPTEMBER 2022

- Task: Speech Transcription on the Bengali Common Voices Dataset
- Model: wav2vec2
- Kaggle: [DL Sprint - BUET CSE Fest 2022](#)
- Team Name: YellowKing, Account: [Salman Khondker](#)
- GitHub: [DLsprint2022-Champion](#)
- Paper: [\[arXiv\]](#)

EEE DAY Datathon 2023

1st Runners Up

Hosted by EEE BUET, Apurba Technologies Ltd.
FEBRUARY - MARCH 2023

- Task: Grammatical Error Detection on an in-house dataset
- Model: T5 Transformer
- Kaggle: [Apurba presents Bhashabhrom: EEE DAY 2023 Datathon](#)
- Team Name: Schadenfreude, Account: [Salman Khondker](#)
- GitHub Link: [Bangla-GED](#)
- Paper: [\[arXiv\]](#)

Robi Datathon 3.0 2024

1st Runners Up

Hosted by Robi Axiata Ltd, Huawei
APRIL - MAY 2024

- Task: Customer Purchase Behavior Prediction
- Model: Random Forest Regressor, XGBoost
- Kaggle Qualifier Round: [Problem 1](#), [Problem 2](#), [Problem 3](#)
- Team Name: YellowKing

AI For Bangla 2023

Honorable Mention

Hosted by EBLICT, Bangladesh Computer Council
DECEMBER 2022 - MARCH 2023

- Task: First word-level Bangladeshi Sign Language Dataset
- Model: 3DCNN, Image-RNN, STGCN
- Host Page: [EBLICT Facebook](#)
- Team Name: Silent Sentinels
- GitHub Link: [BdSL40](#)